Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/01/2005
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Project Title:	STUDIES OF THERMAL RADIATION EFFECTS ON HIGH PRESSURE FLAME DYNAMICS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	08540-5263	**Congressional District:**	12
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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Flight Program:			
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Task Description:	The effect of CO2 addition on flame speed and flammability limit of methane-air and methane-oxygen mixtures will be measured by a spherical propagating and counterflow flames. The spectral radiation will be detected and analyzed by using an infrared spectrometer and the InSb detector. An optimised radiation model will be developed for spectral radiation prediction by comparing the predicted and measured radiation flux at given spectral wavelength. A drop module will be designed and built. 1. Measure the effect of CO2 addition on flame speed and flammability limit of methane-air and methane-oxygen. 2. Measure the the flame spectral radiation by using a spectrometer. and InSb detector. 3. Develop an optimised radiation model 4. Design and build a drop module (rig). 1. Provide the first hand data of the effect of CO2 addition on fire control and		

	burning limit. 2. Establish a computation model for spectral radiation computation. 3. Provide flame radiation database for microgravity experiment
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. Establish a simplified model for fire initation and extinguishment. 2. Enable modeling of fire spreading in microgravity and normal gravity enviroment. 3. Provide fire detection capability using a CO2 radiation sensor.
Task Progress:	Major accomplishments and summaries are listed below: 1. We developed a new pressure release type high pressure spherical flame facility for the measurements of the laminar burning velocity with increased radiation optical thickness. It enables flame experiments in a pressure range between 0.001 and 30 atm. Xiao Qin and Yiguang Ju presented a paper entitled “Measurements of burning velocities of dimethyl ether and air premixed flames at elevated pressures” at the 30th International Symposium of Combustion in Chicago, IL in July 2004. In this paper, dimethyl ether and air flames, which allows high CO2 recycle, was studied. Experiments were conducted at pressures up to 10 atm. The measurement system was validated using laminar burning velocities of methane-air flames. A Comparison with the previous experimental data shows an excellent agreement and demonstrates the accuracy and reliability of the present experimental system. The flame speeds of dimethyl ether and air flames were measured and compared with the predictions using both full and reduced kinetic mechanisms. It was found that flame speed decreases considerably with the increase of pressure and the measured flame speeds are significantly lower than the predictions based on the available kinetic models. These results suggest that the existing kinetic models have to be significantly modified to predict reasonable flame propagation of dimethyl ether flames. Markstein lengths and the overall reaction order at different equivalence ratios were extracted from the flame speed data at elevated pressures. Therefore, the present study yields valuable flame speed data and submodels for high pressure dimethyl ether and air combustion. 2. Yuan Xue and Yiguang Ju presented a paper entitled “Extinction and flame bifurcations of stretched dimethyl-ether premixed flames” at the 30th International Symposium of Combustion in Chicago, IL in July 2004. In this paper, the combined effects of radiation and flame stretch on the extinction and flammability limits in the counterflow flames were studied. The major conclusions are as follows: a) Markstein length of lean dimethyl ether-air premixed is extracted from the counterflow flames using a method based on the flame front. The predicted positive Markstein length agrees well with the experimental data. The results showed that flow stretch significantly reduces the flame speed and narrows the flammability limit of the lean dimethyl ether-air flame. b) The combined effect of radiation and stretch results in a new flame bifurcation and multiple flame regimes for dimethyl ether-air flames. At equivalence ratio slightly higher than the flammability limit of the planar flame, the distant flame regime appears at low stretch rate. With an increase of equivalence ratio, in addition to the distant flame, a weak flame also appears at moderate stretch rates. With a further increase of equivalence ratio, the distant flame and the weak flame merge together which leads to the splitting of the weak flame branch into two new flame branches, a low stretch weak flame and a high stretch weak flame. c) Flame stability analysis demonstrated that the high stretch weak flame is also stable. A K-shaped flammability diagram showing various flame regimes and their flammable regions are obtained. The results demonstrated for the first time that the complicated bifurcations predicted using the one-step chemistry really happen for the DME-air flames. 3. Conducted flame speed measurements of CO2–He diluted methane-oxygen flames at atmospheric and elevated pressures (10 atm) at normal gravity with CO2 up to 50%. The results were compared with predictions with a newly developed spherical flame propagation code. A paper entitled “Experimental and numerical study of spectral dependent radiation on flame propagation” will be presented at the 43rd AIAA Aerospace Sciences Meeting and Exhibit on Jan.10, 2005. 4. Designed and fabricated a pair of nozzle burners and established methane air counterflow flames at normal gravity. The counterflow experiment setup was tested by measuring the extinction limit and temperature of methane air nonpremixed flames. The measured limit and flame temperature excellently matched the results reported in the literature, which validated the system. The spectral radiation properties of CH4-O2-CO2-He counterflow flames will be measured in this system. 5. Preparation for spectral dependent radiation measurements. We have ordered the following optical measurement system from Princeton Instruments for radiation detection: a) Spectrometer (SpectraPro 2500i, 200nm-10um) b) Infrared Detection assembly (InSb) for 1.5 to 5.0 microns c) Lock in amplifier (Stanford Research) 6. Development of code: a) A code package for simulating spherical flame propagation with narrowband radiation model and adaptive grids was developed. Using premixed hydrogen-air flames at normal gravity and pressure, the grid independency and accuracy of the code was tested. b) A spectral narrow band based correlated K model is currently underdevelopment for optimization.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Qin X, Ju Y. "Measurements of burning velocities of dimethyl ether and air premixed flames at elevated pressures." Proceedings of the Combustion Institute 2005 Jan;30(1): 233–40. http://dx.doi.org/10.1016/j.proci.2004.08.251 , Jan-2005
Articles in Peer-reviewed Journals	Ju Y, Xue Y. "Extinction and flame bifurcations of stretched dimethyl ether premixed flames." Proceedings of the Combustion Institute 2005 Jan;30(1):295–301. http://dx.doi.org/10.1016/j.proci.2004.08.258 , Jan-2005

Articles in Peer-reviewed Journals	Guo H, Smallwood GJ, Liu F, Ju Y, Gülder OL. "The effect of hydrogen addition on flammability limit and NOx emission in ultra-lean counterflow CH4/air premixed flames." Proceedings of the Combustion Institute 2005 Jan;30(1):303–11. http://dx.doi.org/10.1016/j.proci.2004.08.177 , Jan-2005
Articles in Peer-reviewed Journals	Yuan J, Ju Y, Law CK. "Coupled hydrodynamic and diffusional-thermal instabilities in flame propagation at subunity Lewis numbers." Physics of Fluids, 2005 Jul; 17(7):074106. http://dx.doi.org/10.1063/1.1964845 , Jul-2005
Articles in Peer-reviewed Journals	Minaev S, Fursenko R, Ju Y, Law CK. "Stability analysis of near-limit stretched premixed flames." Journal of Fluid Mechanics, 2003 Jul;488:225-44. http://dx.doi.org/10.1017/S0022112003004853 , Jul-2003
Books/Book Chapters	Ju Y, Lee JSH. "Flame front." in "Encyclopedia of Nonlinear Science." Ed. A. Scott. Routledge, Taylor & Francis Group, http://www.routledge-ny.com/ref/nonlinearsci/ , Jul-2004